

Cindy Hoffman, D.O.
Dermatology
Residency Director St. Barnabas

Requirements

- Fitzpatrick's
Color Atlas & Synopsis of
Clinical Dermatology
- Handouts/Class notes

Papulosquamous Disorders

- Seborrheic Dermatitis
- Psoriasis Vulgaris
- Pityriasis Rosea
- Lichen Planus
- Cutaneous T Cell Lymphoma
- Fungal Infections

Case 1

- 39 yo male c/o itchy rash of scalp that comes and goes for many years. He reports the rash sometimes affects his face, especially in the eyebrow area. No treatment has been attempted.
- No history of skin disease
- PMHx: neg
- Meds: none

Case 1



Case 1



Case 1



*Elbows and knees unaffected
Nails normal

Differential Diagnosis

- Psoriasis
- Seborrheic dermatitis
- Atopic Dermatitis
- Pityriasis rosea
- Tinea infection

Differential Diagnosis

Seborrheic Dermatitis

- History
 - Chronic, relapsing
- Symptoms
 - itchy
- PE
 - Ill-defined, greasy, yellow, fine white scale
- Distribution
 - Scalp, nasolabial folds, eyebrows, central chest

Psoriasis

- History
 - Chronic, arthritis, joint pains
- Symptoms
 - Usually not itchy
- PE
 - Sharply demarcated, more deeply red, **thicker scale**
 - Nail findings
- Distribution
 - More widespread
 - Typically affects, **scalp, elbows, knees**, other areas
 - Rarely affects face

Seborrheic dermatitis

- Workup
 - HIV testing if new onset and severe
- Treatment
 - Selenium sulfide or zinc pyrithione shampoo
(Selsun Blue, Head and Shoulders)
 - Topical steroids
 - High potency for scalp
 - Low potency for face

Case 2

- 24 yo female presents with rash on her abdomen and back. She states the rash started with **one spot** on her abdomen 1 month ago. 2 weeks ago she noticed more spots began appearing. Minimal pruritus(itching)
- Mild URI 6 weeks prior (untreated)
- MedHx: neg
- Meds: none

Case 2



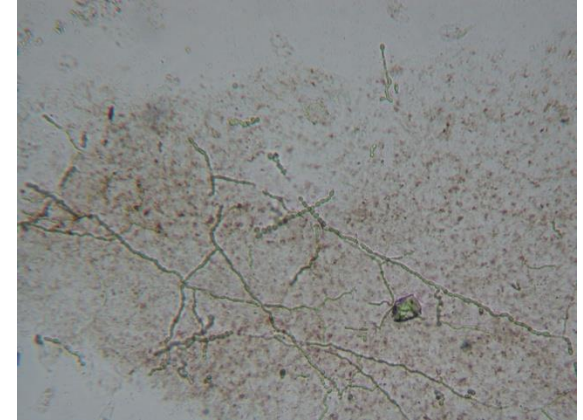
Differential Diagnosis

- Tinea corporis
- Tinea versicolor
- Guttate psoriasis
- Pityriasis rosea
- Drug eruption
- Secondary syphilis

What test can we order to narrow down the differential diagnosis

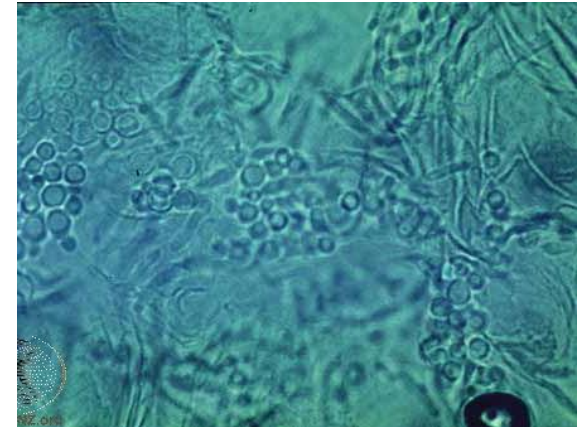
Tinea corporis:

KOH: septate, branching hyphae



Tinea versicolor:

KOH: spaghetti & meatballs



Secondary Syphilis:

RPR / VDRL

Pityriasis rosea and
guttate psoriasis:

ASO titer

Pityriasis Rosea

- Acute exanthematous eruption
- Occurs in spring & fall
- ? Preceding URI (upper respiratory infection)
- PE: **Herald patch** (80% of pts)-a single plaque develops usually on trunk, then
2-21 days later a generalized eruption w/scaling, papules & plaques in **Christmas tree distribution**,
- confined to trunk & proximal extremities, along skin lines
- Clinical course: spontaneous remission in 6 to 12 wks; recurrences uncommon
- TX: symptomatic

Case 3

- 15 yo male develops a generalized rash that started after an episode of strep throat 5 days ago.

Case 3: PE

- multiple, 1 to 2 cm, scattered, scaly, salmon pink papules in a generalized distribution
- spares the palms and soles
- **tear drops**



Case 3: DDX

- Pityriasis rosea
- Drug eruption
- Secondary syphilis
- Tinea corporis
- Guttate psoriasis

Eruptive Guttate Psoriasis

- Rare, appears rapidly
- multiple, 1 to 2 cm, scattered, scaly, salmon pink papules in a generalized distribution, sparing the palms and soles; **tear drops**
- DDx: drug eruption, secondary syphilis, pityriasis rosea, tinea corporis (fungus)
- Tests: ASO titer and throat culture (in those with symptoms of a strep infection)
 - VDRL (syphilis)
 - KOH (tinea)

Differential Diagnosis

Pityriasis rosea

- History
 - Recent URI / strep pharyngitis
- Distribution
 - Trunk and proximal extremities
 - Christmas tree pattern
 - Herald patch

Guttate psoriasis

- History
 - Recent URI / strep pharyngitis
- Distribution
 - Generalized, may be concentrated over knees and elbows
- Lesion Morphology
 - Tear Drop-like

**Both may be associated with antecedent streptococcal pharyngitis and +ASO titer*

DDX

PR



Guttate Psoriasis



Case 4

- 22 yo male with rash of trunk x 3 weeks, since the beginning of summer. He reports the rash was present last summer as well, but it seemed to go away during the winter.



Sharply demarcated scaly hyperpigmented macules and patches



Differential Diagnosis

- Tinea versicolor
- Pityriasis rosea
- Secondary syphilis
- Drug eruption
- psoriasis

Tinea Versicolor

- Lesions are asymptomatic w/ minimal pruritus, lasts months to years
- PE: sharply marginated, hyper- or hypopigmented, scaly, round or oval macules of varying size
upper trunk, arms, neck, axillae, abdomen
- Dx: KOH-spaghetti and meatballs
- Tx: antiyeast medication; selenium sulfide





Case 5

- A 27 year old female develops a rash on her **scalp, elbows, knees**, lower legs, ankles, and lower back. The lesions are covered with a **thick white scale**. Attempts to remove the scale by scratching causes the lesions to bleed.
- Her grandmother had a similar rash and severe arthritis



Differential Diagnosis

- Psoriasis
- Tinea corporis
- Secondary syphilis
- Contact dermatitis
- Atopic dermatitis

Psoriasis Hints

- Clinical-thick silvery white scaly plaques
- Distribution
 - Extensor surfaces, scalp
- Family history
- Joint pain

Psoriasis Hints

The Koebner Phenomenon

- also known as isomorphic response
- refers to the induction of new psoriasis skin lesions following local trauma or injury to the skin
- Triggers include:
 - Physical trauma (cuts, abrasions)
 - acupuncture or tattoo needles
 - Mechanical (constant pressure and rubbing) - elbows, knees
 - Allergic (contact dermatitis)
 - Chemical (burns)

The Koebner Phenomenon



Psoriasis lesions in tattooed skin



Psoriasis lesion induced after
Herpes Zoster outbreak

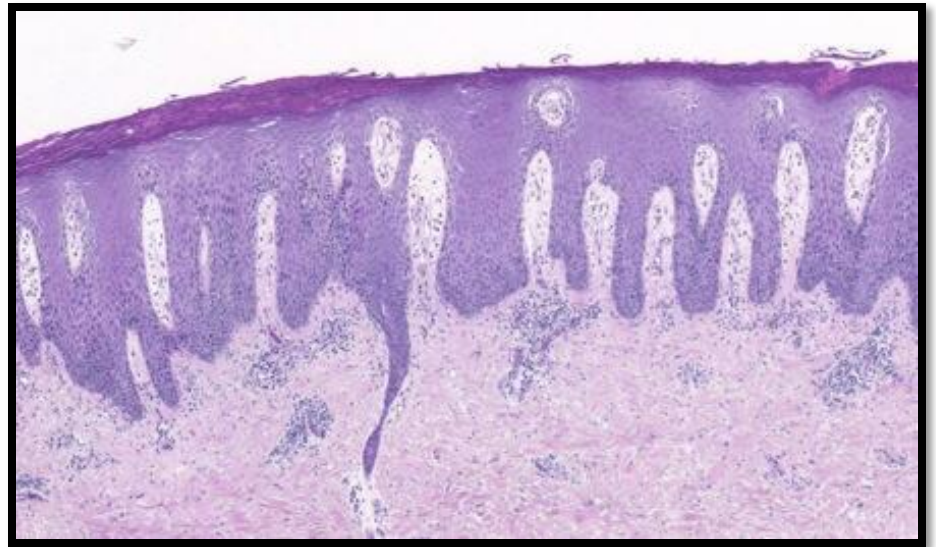


Psoriasis lesions from surgical incision

Psoriasis Hints

Auspitz sign

- When scale of psoriasis is scraped, pinpoint bleeding occurs
- These bleeding points represent the dilated and tortuous capillaries in the papillary dermis with thinning of overlying suprapapillary portions of epidermis in psoriasis plaques.



Psoriasis Hints

Nail Findings

- Onycholysis on right
- Pitting below



- Oil spot below
(pathognomonic for psoriasis)



Case 6

- 54 yo Caucasian male presents with asymptomatic rash of trunk that comes and goes for years.
- Tried several prescription creams, some of which seem to help temporarily, but the rash keeps returning.



Confluent erythematous scaly patches and plaques with irregular configuration.

DDx

- Psoriasis
- Tinea corporis
- Contact dermatitis (irritant or allergic)
- Pityriasis rosea (atypical)
- Mycosis Fungoides

Mycosis Fungoides

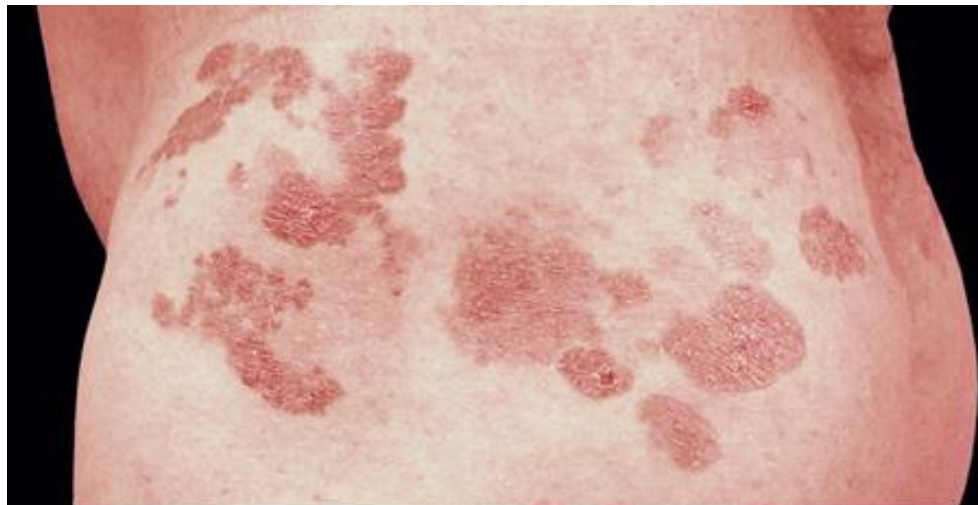
- MF is the most common cutaneous lymphoma.
- Occurs in mid to late adulthood (age 55-60) with male predominance of 2:1.
- A clonal **proliferation** of **T cells**
- Categorized as patch, plaque, or tumor stage.
- Prognosis related to stage.
- Treatment: symptom-oriented and stage-dependent.
- Etiology unknown.

MF Clinical Manifestations

- For months to years, often preceded by various diagnoses such as psoriasis, nummular dermatitis, and tinea corporis, parapsoriasis
- Symptoms: +/- pruritus
- Skin lesions are classified into patches, plaques, and tumor stage. Patients may have simultaneously more than one type of lesions.

Patches

- Randomly distributed, scaling or nonscaling patches in different shades of red
- Well- or ill-defined
- at first superficial, much like eczema or psoriasis or mimicking dermatophytosis ("mycosis"), and later becoming thicker.



Plaques

- Round, oval, annular, and of bizarre configuration
- Lesions are randomly distributed but in early stages often spare exposed areas.



Tumors

- Later lesions consist of nodules and tumors, with or without ulceration.



Case 7

Whats the diagnosis?

The 5 P's

- purple
- polygonal
(plane figure with multi sides)
- pruritic (itchy)
- planar (flat)
- papules



Lichen Planus

Where else should you look if you suspect LP?



What is this called?
Wickham's striae

What should we ask about in the
patients medical history ?

Do you want to order any tests?

Hepatitis C

Case 8

- A 30-year-old female with two pet dogs at home develops a rash on her left arm.
- The rash is red and scaly, and the lesions are erythematous, annular and have a well-defined edge, with ***central clearing***



Differential Diagnosis

- Contact dermatitis
- Psoriasis
- Pityriasis rosea
- Seborrheic dermatitis
- Tinea corporis

Tinea corporis:

What is the most common cause?

- ***Trichophyton rubrum*** is the most common cause
- Dermatophyte infection via autoinoculation or animal contact
- PE: small to large, scaly, sharply marginated **annular plaques** w/or w/o pustules or vesicles at margin, **central clearing**

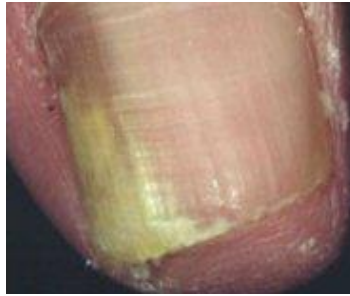
What is tinea incognito?

- The presentation of dermatophytosis after application of topical steroids



Onychomycosis

- distal / lateral subungual



- *T. rubrum*

- superficial white



- *T. mentagrophytes*

- proximal subungual



- *T. rubrum*
- HIV associated

Case 9

A patient presents with a rash of the inguinal area...What is the differential diagnosis?



- Intertrigo
- Tinea cruris
- Inverse psoriasis
- Candidiasis
- Erythrasma
- Eczematous

How can we distinguish the etiology?

Intertrigo

- Occurs in places where opposing skin surfaces rub against each other: axillae, creases of lips, inguinal, intergluteal creases, and inframammary area
- Well-demarcated, erythematous patches or plaques that oppose each other on either side of the skin fold (mirror image)
- Diagnosis of exclusion
 - Contact dermatitis, psoriasis, fungal (KOH / culture)

Intertrigo



Tinea cruris- (jock itch)

- upper inner thighs and groin, most often in postpubertal males
- Caused by the dermatophyte ***Trichopyton rubrum***
- PE: bilateral **annular** plaques with advancing border, **central clearing**
- **spares the scrotum and penis
 - vs. candidasis and eczematous etiologies
- Dx: KOH examination or fungal culture

Tinea cruris



Courtesy of
The Geraldine Kaminski Medical Mycology Library
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Inverse psoriasis

- Psoriasis localized to intertriginous areas: axillae, inframammary area, perineum, inguinal creases
- PE: deep pink to red color and well-defined borders
- **often involves penis and scrotum
- Dx: psoriasis elsewhere on body
negative KOH

Inverse Psoriasis



Source: McPhee SJ, Papadakis MA: *Current Medical Diagnosis and Treatment* 2011, 50th Edition: <http://www.accessmedicine.com>

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Candidiasis

- Infection with *Candida* species
- Common in elderly, diabetic, immunocompromised patients
- Occur in skin folds, where occlusion from clothing produces warm, moist conditions
- PE: **beefy red** color to lesions with **satellite pustules** seen beyond the border
maceration and fissures may be present
- Dx: KOH: budding yeast or + culture for *Candida*

Candidiasis



Source: Wolff K, Johnson RA: *Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology*, 6th Edition: <http://www.accessmedicine.com>

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Erythrasma

- bacterial infection caused by *Corynebacterium minutissimum*.
 - part of the normal skin flora which proliferates in affected areas.
- Predisposing factors include excessive sweating, obesity, diabetes, poor hygiene, advanced age and other conditions which affect immunity.
- Diagnosis can be made by shining a Wood's light on the affected areas;
- a coral-red fluorescence caused by porphyrins produced by the bacteria.

Woods Lamp Examination uses UV-A to detect bacteria



Eczematous

- Lichen simplex chronicus: often solitary, pruritic eczematous eruption exacerbated and caused by **repetitive rubbing and scratching**
- Most often occurs in adults with history of atopic dermatitis
- PE: focal **lichenified plaques** (thickened) that are poorly demarcated
- Dx: KOH neg

Lichen Simplex Chronicus of scrotum



Source: Tintinalli JE, Stapczynski JS, Ma OJ, Cline DM, Cydulka RK, Meckler GD:
Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 7th Edition:
<http://www.accessmedicine.com>
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Case 10

A 67 year old African American female presents to the dermatology clinic with a complaint of a “red, oozing lesion on the inner aspect of her right ankle” that she claimed began as a “rash” a few months prior but did not seek medical attention until she noticed recently the skin had become raw and appeared infected. She also complains her lower legs are constantly swollen. The patient has a medical history of hypertension and was diagnosed with a deep vein thrombosis last year.

On physical exam you note an obese elderly female with bilateral edema in her lower extremities, and a weeping erythematous ulcer with hyperpigmented borders on the medial aspect of her right ankle, as seen in the image provided



- Aka stasis eczema
- Common, **inflammatory** dermatosis of **lower extremities**
- Most common in patients with **chronic venous insufficiency**
 - Risk factors: age, family hx, female sex, standing occupation, obesity, hx of DVT, CHF, HTN
 - Stasis dermatitis is a **late manifestatoin of chronic** venous disease
- Associated with **varicose veins**, dependent chronic **edema**, **hyperpigmentation**, and **ulcerations**
 - (Chronic venous insufficiency → edema, hyperpigmentation, eczema, fibrosis, atrophy, and ulceration)
- Management: treat underlying venous HTN



- Early signs: **erythema**, **scaling**, and slight **hyperpigmentation** in the **medial supramalleolar** region.
- Increased **scaling**, **peripheral edema**, **erosions**, **crusts**, and secondary **bacterial infection**
- Advanced stasis dermatitis with **hyperpigmentation**, **lichenification**, **sclerosis**, and **ulceration**.



- <https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>